MOHAMMAD NAVED

AI-Assisted Developer | Prompt Engineer | Full-Stack Product Builder

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PROFESSIONAL SUMMARY

AI-assisted developer who designs, ships, and tests full-stack, IoT, and LLM-powered products by directing autonomous AI coding agents and workflows. Hands-on experience integrating multi-model LLM APIs (Claude, Gemini, NVIDIA NIM, GLM), prompt engineering, natural-language-to-SQL, and cloud deployment. Owns the product architecture, prompt specifications, testing gates, and edge-case debugging; directs AI agents for rapid, clean code generation. Shipped a Google-certified smart home system (71/75 test cases passed), a multi-tenant cloud pharmacy SaaS, and an AI analytics platform. B.Tech in Computer Science.

TECHNICAL SKILLS

AI and LLM: Multi-LLM APIs (Claude, Gemini, NVIDIA NIM, GLM), Prompt Engineering, Agentic Workflows, Natural-Language-to-SQL, AI Coding Agents & Agentic Workflows

Web Technologies: React 19, Next.js, TypeScript, JavaScript, Tailwind CSS v4, Vite

Backend and Data: FastAPI, Node.js, REST APIs, Supabase, PostgreSQL, SQL, DuckDB, Python, Power BI

IoT and Cloud: ESP32, MQTT, AWS IoT Core, AWS App Runner, Docker, Vercel, GitHub Actions

EXPERIENCE

Codtrex AI — Boutique AI Engineering Studio

2025 - Present

Founder and AI Engineer • codtrex.vercel.app • React 19, Vite, Tailwind CSS, Supabase, Vercel

- Run a boutique AI engineering studio offering multi-provider LLM integration (Claude, Gemini, NVIDIA NIM, GLM), full-stack web products, and IoT systems for clients with 14-20 day rapid turnaround.
- Built and deployed the studio website with a Gemini-powered AI chat assistant, service tiers, and interactive case studies.
- Deliver client projects by designing system requirements, writing structured prompts for AI coding agents, reviewing output, and testing before production deployment.

FEATURED PROJECTS

4Layers — Smart Home IoT Ecosystem

2025 - Present

ESP32, AWS IoT Core, MQTT, React Native, FastAPI, Docker, Google Home API

- Passed 71 of 75 tests in the Google Home Test Suite for official Google Smart Home certification; devices listed and working in the Google Home app.
- Designed 3-source state synchronization (ESP32 hardware, AWS IoT Core MQTT, Google Home) so physical switches, mobile app, and voice commands stay consistent.
- Diagnosed and resolved a certification-blocking issue where Google Home reverted local device states, by adding a QA Hold Mode with automated TTL.
- Implemented OTA firmware updates with version pinning and dual-partition rollback protection for deployed devices.
- Designed 4-channel relay switch boards and multi-speed fan regulators; provisioned per-device mTLS certificates through AWS IoT Core.

MediStock — Cloud Pharmacy Management SaaS

2025 - Present

React 18, Supabase, PostgreSQL, Capacitor Android, Vercel • medistock-pos.vercel.app

- Shipped a B2B SaaS for inventory management, FEFO batch allocation, expiry tracking, billing, and supplier accounting.
- Designed a multi-tenant PostgreSQL schema with Row Level Security (RLS) to ensure complete data isolation across pharmacies.
- Deployed as a live web application and an Android APK via Capacitor; preparing Google Play Store submission.

PROJECTS (CONTINUED)

DataLens AI — Natural-Language-to-SQL Analytics Platform *2025*

React, TypeScript, Recharts, Gemini API, DuckDB, SQL

- Built an AI analytics platform with a chat interface that converts plain English questions into SQL so non-technical users can query CSV and Excel data.
- Added an ML Lab module that explains model outputs and surfaces business insights using Gemini prompt templates and schema injection.
- Designed the end-to-end data pipeline, SQL engine execution safety (read-only queries), and visualization layer.

UsedTech Market — C2C Electronics Marketplace *2025*

Next.js, React 19, TypeScript, Supabase, Tailwind CSS

- Designed a marketplace for used electronics with real-time listings, advanced search filters, and seller dashboards.
- Defined the product data model, authentication flow, and complete listing lifecycle (draft, published, sold).
- Implemented Supabase real-time subscriptions for live listing status and seller notification alerts.

Interactive Portfolio & Automation Pipeline *2025*

React, TypeScript, Tailwind CSS, Node.js, Puppeteer, Gemini API • mohammad-naved-portfolio.vercel.app

- Built a responsive, dark/light mode portfolio featuring an interactive Gemini-powered AI chatbot assistant.
- Automated resume PDF generation from HTML source using a Node.js and Puppeteer pipeline for pixel-perfect PDF rendering.

AI ENGINEERING METHODOLOGY

- **Contract & Schema First:** Define strict data contracts, TypeScript interfaces, and PostgreSQL schemas before prompting AI agents, eliminating API hallucinations.
- **Automated Verification:** Direct agents to generate unit and integration test suites; execute automated test suites (e.g., Google Home 75-test certification suite) before code merge.
- **Deterministic Failsafes:** Enforce read-only locks, database row-level security (RLS), and in-memory execution boundaries (DuckDB) for high-risk operations.

EDUCATION

B.Tech, Computer Science — RGPV University, Indore *Graduated: 2025*

Rajiv Gandhi Proudyogiki Vishwavidyalaya • Madhya Pradesh, India

CERTIFICATIONS & ACHIEVEMENTS

- **Google Smart Home Certification:** 4Layers IoT Ecosystem (71 of 75 tests passed in official Google Home Test Suite).
- **Generative AI with Gemini:** Prompt Engineering and LLM Application Development.
- **Python for Data Analysis and Visualization:** Exploratory data analysis, Pandas, and visualization pipelines.
- **SQL for Data Analytics:** Multi-table relational queries, joins, window functions, and schema normalization.
- **Power BI:** Interactive dashboard design, DAX data modeling, and business reporting.